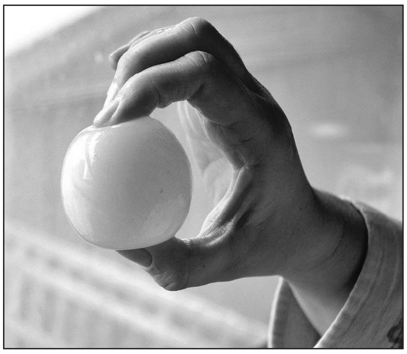
Examiner only

5. A hen's egg is a single cell which is surrounded by a calcium carbonate shell.
Six hen's eggs were soaked in 4% ethanoic acid for 24 hours to dissolve their shells. After this time each egg is still intact as it is surrounded by a cell membrane. The initial masses of the 'naked eggs' (eggs without shells) were recorded.

'naked egg'



The six 'naked eggs' were each placed in sucrose (sugar) solutions of different concentrations and left for 24 hours. After this time the final masses of the 'naked eggs' were recorded.

Results table

Egg number	Concentration of sucrose solution (%)	Initial mass (g)	Final mass (g)	Change in mass (g)	% change in mass
1	0	61.5	67.2	5.7	9.3
2	5	64.3	68.9	4.6	7.2
3	10	59.8	63.2	3.4	5.7
4	15	60.7	59.8	–0.9	–1.5
5	20	62.6	59.9	–2.7	–4.3
6	25	60.9	56.1	–4.8	....

(a) **Complete the table** above by calculating the % change in mass for egg number **6**. [2]
Space for working.

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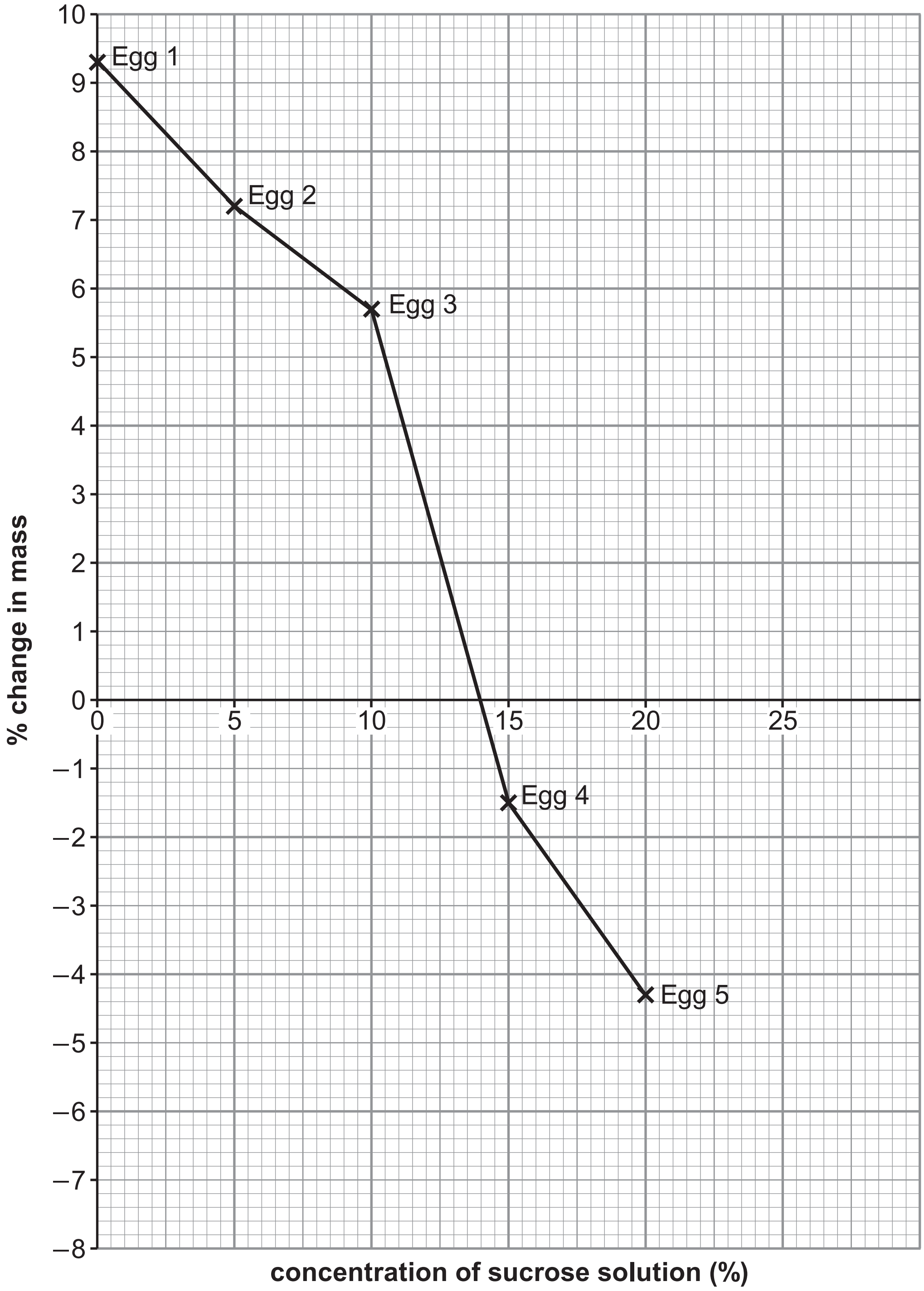
(b) In the graph below the % change in mass of the 'naked eggs' is plotted against the concentration of sucrose solution.



(i) **Complete** the **graph** by adding the percentage change in mass for egg number **6**. [1]

(ii) From the graph state the concentration of sucrose solution where there was no net movement of water. [1]

concentration of sucrose solution = ... %

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(c) Explain the results for egg number **3** and egg number **5**. [4]

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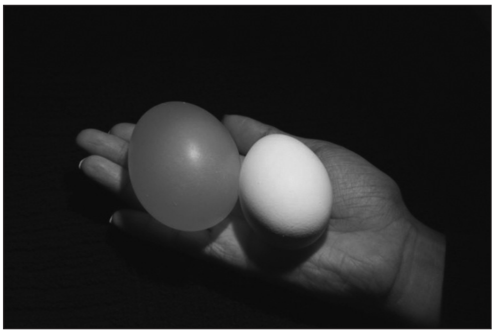
(d) The photograph below shows a complete hen's egg on the right and one of the 'naked eggs', used in this experiment, on the left.

Suggest a concentration of sucrose solution in which this 'naked egg' was kept. [1]



suggested concentration of sucrose solution = ... %

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